

A5 - Design for Strength and Stiffness I

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MEGR 2157
Time spent: approximately **5 hours**

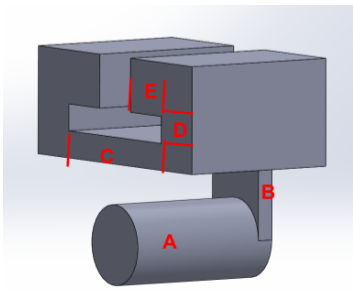
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Trouble reading the formatting? View all original handwritten work — full-page calculations and both drawings. The same scans are included at the end of the submission PDF, with a clickable link to them.

Objective and design decisions

Design a bracket using five strength analyses and five stiffness analyses, then design the additional MEGR 2157 linkage and its fits. Each feature must meet a safety factor of 4 and a local deflection limit of 0.005 in. This report follows the updated assignment, which requires two paper multiview sketches and a PDF submission; it does not require CAD files.

I chose a symmetric bracket with a centered connector B and two opposite pin overhangs A. Each strap leg applies 600 lbf, so the bracket carries 1,200 lbf total. The linkage case replaces the strap with two identical links, each carrying 600 lbf. These are alternative load cases, not simultaneous loads.



Assignment-provided A–E feature diagram. My design uses a centered B connector with symmetric pin overhangs. .

The load path is $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow$ rigid T-beam. Opposing A moments cancel at B. Each upper lip E carries 600 lbf and transfers a 900 lbf·in moment through its sidewall D into C. Consequently, C includes end couples in addition to the center load suggested in the assignment hint. All sketches use a rotated page orientation for convenience; gravity is neglected.

Given values, chosen dimensions, and assumptions

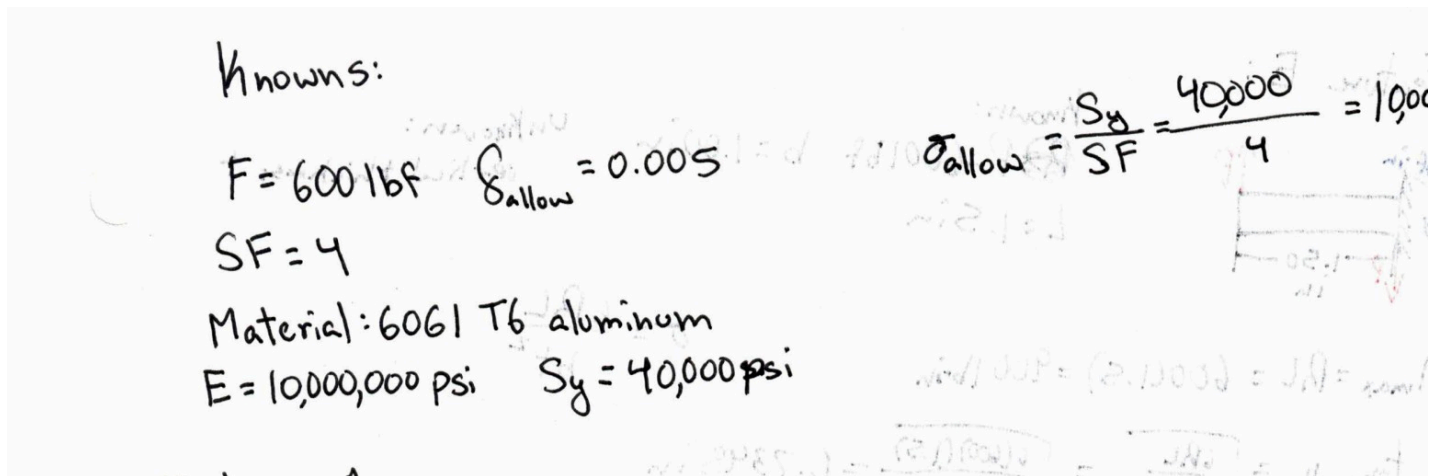
Quantity	Value	Basis
Force per strap leg, F	600 lbf	Chosen within $500 < F < 800$
Total bracket force, P	1,200 lbf	Two symmetric legs
Required safety factor	4	Assignment

Quantity	Value	Basis
Allowable local deformation	0.005 in	Assignment, each feature
Material	Aluminum 6061-T6	Permitted choice
Elastic modulus, E	10,000,000 psi	Rounded typical property [2]
Yield strength, S_y	40,000 psi	Typical property [2]
Allowable normal stress	10,000 psi	$S_y / 4$

The selected material values are typical reference values for this classroom analysis, not a material certificate. All lengths below are in inches unless otherwise stated.

Assumptions: small linear-elastic deformation, symmetric loading, rigid T-beam, no self-weight, no direct-shear failure, and negligible shear deformation. Each stiffness check measures member deformation relative to its connection; it is not the displacement of the complete flexible frame. Nominal section stresses are used, excluding local fillet/hole stress concentrations and press-fit residual stresses. Connections transfer the forces and moments specified in the calculations below. The handwritten FBDs appear within my scanned work; arrow-direction corrections are recorded in the review notes.

Common calculation: $\sigma_{\text{allow}} = S_y / SF = 40,000 / 4 = 10,000 \text{ psi}$.



Original handwritten inputs. The typed values above restore text clipped at the scan edge. .

A - Pin overhang: strength and stiffness

Known: $W = 600 \text{ lbf}$ per overhang, $L = 1.000$, $w = W/L = 600 \text{ lbf/in}$. **Unknown:** diameter d from strength and stiffness.

Assumption: one fixed-root circular cantilever under uniform strap load; the other overhang is its mirror.

Strength

$$R = W = 600 \text{ lbf}; M = WL/2 = 600(1)/2 = 300 \text{ lbf}\cdot\text{in}.$$

$$I = \pi d^4/64; \sigma = Mc/I = 32M/(\pi d^3).$$

$$d_{\text{strength}} = [32M/(\pi \sigma_{\text{allow}})]^{1/3} = [32(300)/(\pi \times 10,000)]^{1/3} = \mathbf{0.6736 \text{ in}}.$$

Stiffness

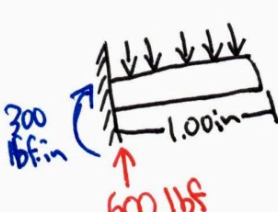
$$\delta = WL^3/(8EI) = 8WL^3/(E\pi d^4).$$

$$d_{\text{stiffness}} = [8WL^3/(E\pi \delta_{\text{allow}})]^{1/4}$$

$$= [8(600)(1)^3/(10^7 \pi \times 0.005)]^{1/4} = \mathbf{0.4181 \text{ in}}.$$

Select nominal diameter **0.750 in**; the final RC3 shaft limits are checked below.

Feature A:



Known:
 $W = 600 \text{ lbf}$, $L = 1.00 \text{ in}$

$$M_{\text{max}} = \frac{WL}{2} = \frac{600(1)}{2} = 300 \text{ lbf}\cdot\text{in}$$
$$\delta = \frac{WL^2}{8EI} = \frac{8WL^3}{E\pi d^4}$$
$$d_{\text{stiffness}} = \sqrt[4]{\frac{8WL^3}{E\pi \delta_{\text{allow}}}} = \sqrt[4]{\frac{8(600)(1)^3}{10^7 \pi (0.005)}} = 0.4181 \text{ in}$$
$$I = \frac{\pi d^4}{64}, \sigma = \frac{M_c}{I} = \frac{32M}{\pi d^3}$$
$$d_{\text{strength}} = \sqrt[3]{\frac{32M}{\pi \sigma_{\text{allow}}}} = \sqrt[3]{\frac{32(300)}{\pi 10000}} = 0.6736 \text{ in}$$

nominal $d = 0.750 \text{ in}$

My handwritten A calculations and FBD. Typed equations and review notes clarify the original arrow direction and clipped text. .

B - Center connector: strength and stiffness

Known: $P = 1,200 \text{ lbf}$, $L = 1.000$, section breadth $b = 0.500$. **Unknown:** thickness t . **Assumption:** centered axial tension; symmetric A moments cancel.

Strength

$$A = bt; \sigma = P/(bt).$$

$$t_{\text{strength}} = P/(b\sigma_{\text{allow}}) = 1,200/(0.500 \times 10,000) = \mathbf{0.2400 \text{ in}}.$$

Stiffness

$$\delta = PL/(Ebt).$$

$$t_{\text{stiffness}} = PL/(Eb\delta_{\text{allow}}) = 1,200(1)/(10^7 \times 0.500 \times 0.005) = \mathbf{0.0480 \text{ in.}}$$

Select $t = \mathbf{0.250 \text{ in}}$ and $b = 0.500 \text{ in}$. In the front view, thickness t is horizontal and b is the depth into the page.

Feature B:

Diagram: A vertical rectangular member with a central downward load of 1200 lbf and two upward reaction loads of 1200 lbf at the top and bottom. The height is labeled as 1.00 in.

Knowns:
 $P = 1200 \text{ lbf}$
 $b = 0.500 \text{ in}$
 $L = 1.00 \text{ in}$

Unknown:
 thickness: t

Strength:
 $A = bt$, $\sigma = \frac{P}{bt}$
 $t_{\text{strength}} = \frac{P}{b\sigma_{\text{allow}}} = \frac{1200}{0.5(10000)} = 0.24 \text{ in}$

Stiffness:
 $\delta = \frac{PL}{Ebt}$
 $t_{\text{stiffness}} = \frac{PL}{Eb\delta_{\text{allow}}} = \frac{1200(1)}{10^7(0.5)(0.005)} = 0.0480 \text{ in}$
 select $t = 0.250 \text{ in}$

Original handwritten B strength and stiffness calculations. .

C - Bottom crossmember: strength and stiffness

Known: center load $P = 1,200 \text{ lbf}$; support-center span $L = 3.500$; depth $b = 1.000$; end reaction 600 lbf ; end-couple magnitude $M_0 = 600(1.500) = 900 \text{ lbf}\cdot\text{in}$. **Unknown:** vertical thickness t . **Assumption:** simply supported member with the center load and opposite end couples transferred through D. The support model permits end rotation; the applied couples are retained explicitly.

Strength

For the left half, $0 \leq x \leq 1.750$: $M(x) = -900 + 600x$.

$M(0) = -900$; $M(1.750) = 150$; therefore $|M|_{\text{max}} = 900 \text{ lbf}\cdot\text{in}$.

$$I = bt^3/12; \sigma = 6|M|_{\text{max}}/(bt^2).$$

$$t_{\text{strength}} = \sqrt{[6|M|_{\text{max}}]/(b\sigma_{\text{allow}})} = \sqrt{[6(900)]/(1 \times 10,000)} = \mathbf{0.7348 \text{ in.}}$$

Stiffness

Taking downward displacement as positive:

$$EI_y = Px(3L^2 - 4x^2)/48 - M_0x(L - x)/2$$

$$= -656.25x + 450x^2 - 100x^3.$$

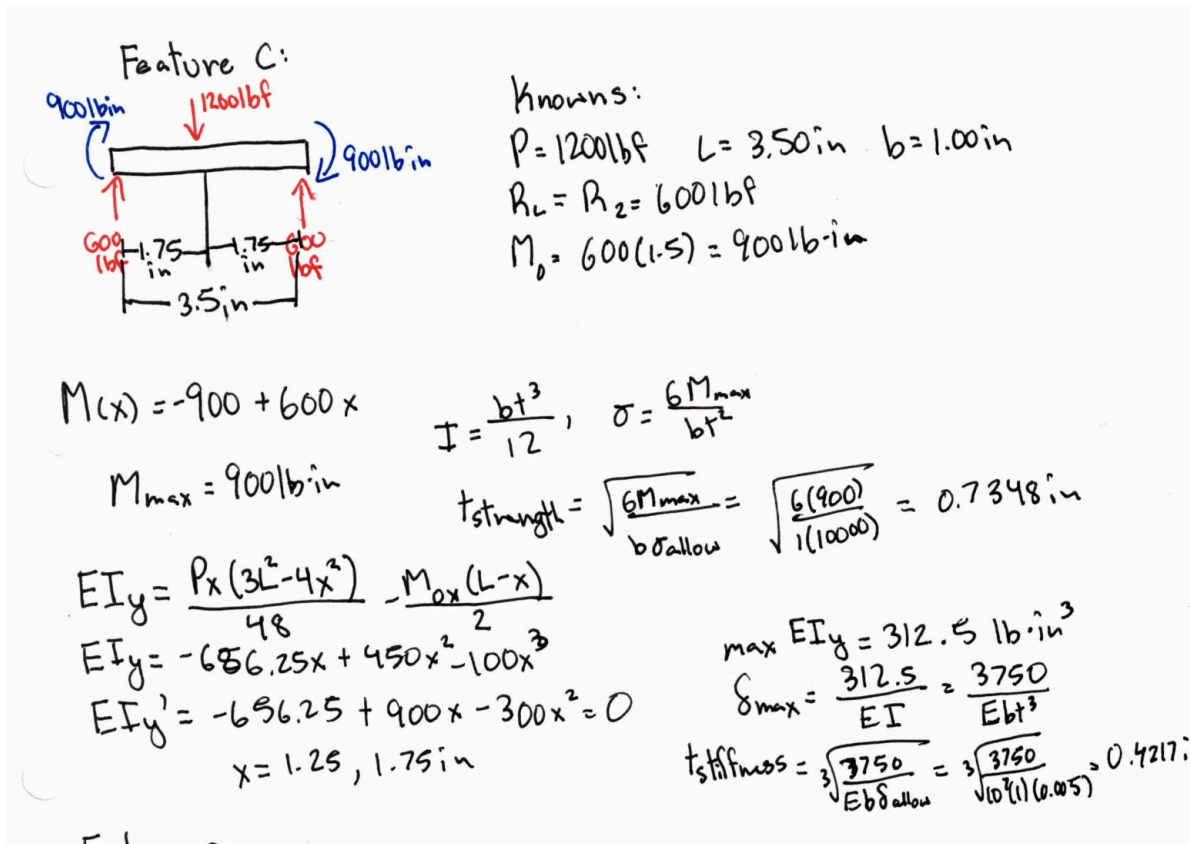
$$EI_y' = -656.25 + 900x - 300x^2 = 0, \text{ giving } x = 1.250 \text{ and } 1.750 \text{ in.}$$

Checking those positions and the end gives $\max |EI_y| = 312.5 \text{ lbf} \cdot \text{in}^3$.

$$\delta_{\max} = 312.5/(EI) = 3,750/(Ebt^3).$$

$$t_{\text{stiffness}} = [3,750/(Eb\delta_{\text{allow}})]^{1/3} = [3,750/(10^7 \times 1 \times 0.005)]^{1/3} = \mathbf{0.4217 \text{ in.}}$$

Select $t = \mathbf{0.750 \text{ in.}}$



Original handwritten C analysis, including the bending-moment and deflection equations. .

D - Sidewall: strength and stiffness

Known: $N = 600 \text{ lbf}$, $M = 900 \text{ lbf} \cdot \text{in}$, centerline height $H = 2.250$, front-view width $h = 1.000$. **Unknown:** depth into the page t . **Assumption:** axial tension plus constant bending; local translation is measured relative to the lower connection with its rigid-body motion removed.

Strength

$$A = ht; I = th^3/12.$$

$$\sigma_{\max} = N/(ht) + 6M/(th^2).$$

$$t_{\text{strength}} = (N/h + 6M/h^2)/\sigma_{\text{allow}}$$

$$= [600/1 + 6(900)/1^2]/10,000 = \mathbf{0.6000 \text{ in.}}$$

Stiffness

$$\delta_{\text{axial}} = NH/(Eht) = 0.000135/t.$$

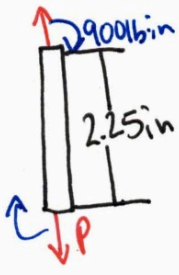
$$\delta_{\text{lateral}} = MH^2/(2EI) = 6MH^2/(Eth^3) = 0.00273375/t.$$

$$\delta = \sqrt{(\delta_{\text{axial}})^2 + (\delta_{\text{lateral}})^2}.$$

$$t_{\text{stiffness}} = \sqrt{(0.000135^2 + 0.00273375^2)}/0.005 = \mathbf{0.5474 \text{ in.}}$$

Select $t = \mathbf{1.000 \text{ in}}$ to match the depth of C and E. The analysis-only paper sketches use the calculated smaller D depths. Their actual centerline heights are shorter than 2.250 in, so retaining this calculation is conservative for those sketches.

Feature D:



$N = 600 \text{ lbf}$ $H = 2.25 \text{ in}$
 $M = 900 \text{ lbf} \cdot \text{in}$ $h = 1.00 \text{ in}$

$A = ht$, $I = \frac{th^3}{12}$ $\sigma_{\max} = \frac{N}{ht} + \frac{6M}{th^2}$

$t_{\text{strength}} = \frac{N/h + \frac{6M}{h^2}}{\sigma_{\text{allow}}} = \frac{600/1 + \frac{6(900)}{1^2}}{10000} = 0.6000 \text{ in}$

$\delta_{\text{axial}} = \frac{NH}{Eht} = \frac{600 \cdot 2.25}{10000 \cdot 1 \cdot 1} = 0.000135$

$\delta_{\text{lateral}} = \frac{MH^2}{2EI} = \frac{6MH^2}{Eth^3} = \frac{0.0027337}{1}$

$\delta = \sqrt{\delta_{\text{axial}}^2 + \delta_{\text{lateral}}^2}$

$t_{\text{stiffness}} = \frac{\sqrt{(0.000135)^2 + (0.0027337)^2}}{0.005} = 0.5474 \text{ in}$

Original handwritten D combined-stress and resultant-deformation calculations. .

E - Upper lip: strength and stiffness

Known: reaction $R = 600$ lbf, effective length $L = 1.500$ from sidewall centerline to inner tip, depth $b = 1.000$. **Unknown:** vertical thickness t . **Assumption:** tip-loaded cantilever; placing the T-beam reaction at the inner tip conservatively maximizes its lever arm.

Strength

$$M = RL = 600(1.500) = 900 \text{ lbf}\cdot\text{in.}$$

$$\sigma = 6RL/(bt^2).$$

$$t_{\text{strength}} = \sqrt{[6RL/(b\sigma_{\text{allow}})]} = \sqrt{[6(600)(1.500)/(1 \times 10,000)]} = \mathbf{0.7348 \text{ in.}}$$

Stiffness

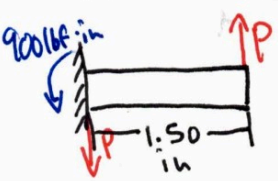
$$\delta = RL^3/(3EI) = 4RL^3/(Ebt^3).$$

$$t_{\text{stiffness}} = [4RL^3/(Eb\delta_{\text{allow}})]^{1/3}$$

$$= [4(600)(1.500)^3/(10^7 \times 1 \times 0.005)]^{1/3} = \mathbf{0.5451 \text{ in.}}$$

Select $t = \mathbf{0.750 \text{ in.}}$

Feature E:



known:
 $P = 600 \text{ lbf}$
 $b = 1.00 \text{ in}$
 $L = 1.5 \text{ in}$

unknown:
 vertical thickness t

$$\sigma = \frac{6RL}{bt^2}$$

$$M_{\text{max}} = RL = 600(1.5) = 900 \text{ lbf}\cdot\text{in}$$

$$t_{\text{strength}} = \sqrt{\frac{6RL}{b\sigma_{\text{allow}}}} = \sqrt{\frac{6(600)(1.5)}{1(10,000)}} = 0.7348 \text{ in}$$

$$t_{\text{stiffness}} = \sqrt[3]{\frac{4RL^3}{Eb\delta_{\text{allow}}}} = \sqrt[3]{\frac{4(600)(1.5)^3}{10^7(1)(0.005)}} = 0.5451 \text{ in}$$

Original handwritten E strength and stiffness calculations. .

Comparison and final design selection

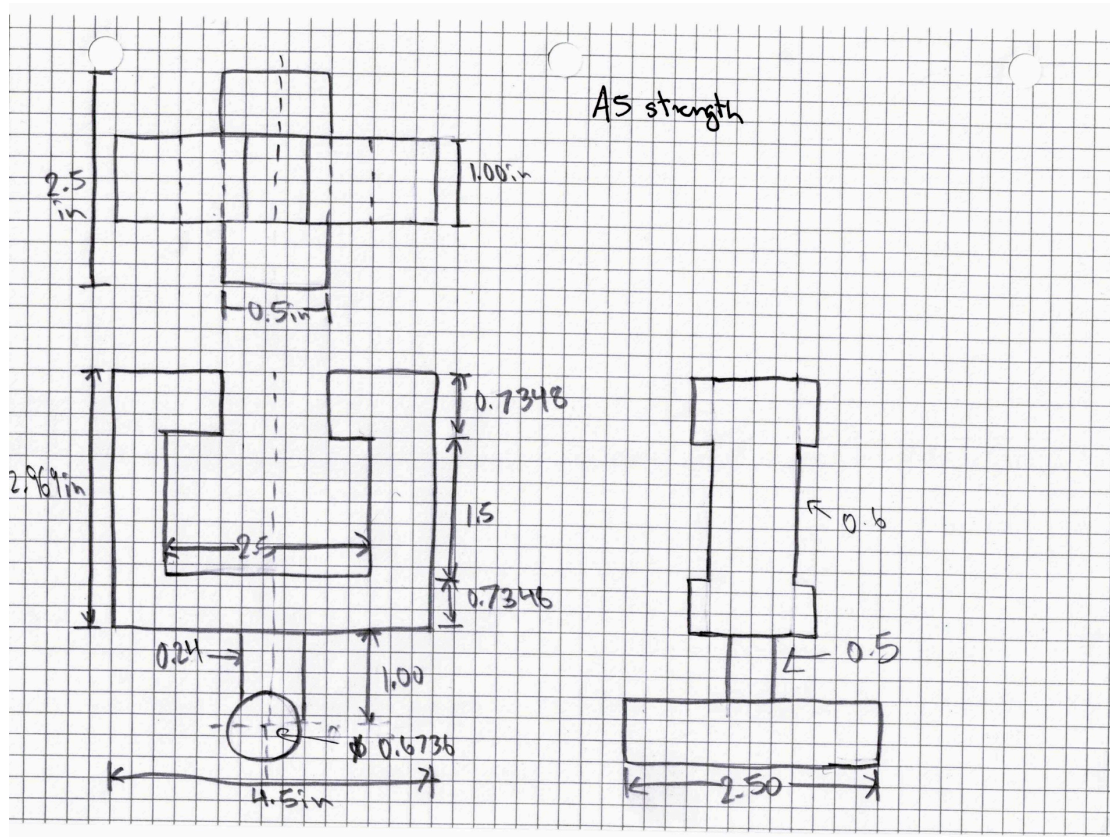
Sized dimension	Strength result (in)	Stiffness result (in)	Selected dimension (in)
A diameter	0.6736	0.4181	0.750 nominal; RC3 limits below

Sized dimension	Strength result (in)	Stiffness result (in)	Selected dimension (in)
B thickness	0.2400	0.0480	0.250
C thickness	0.7348	0.4217	0.750
D depth	0.6000	0.5474	1.000
E thickness	0.7348	0.5451	0.750

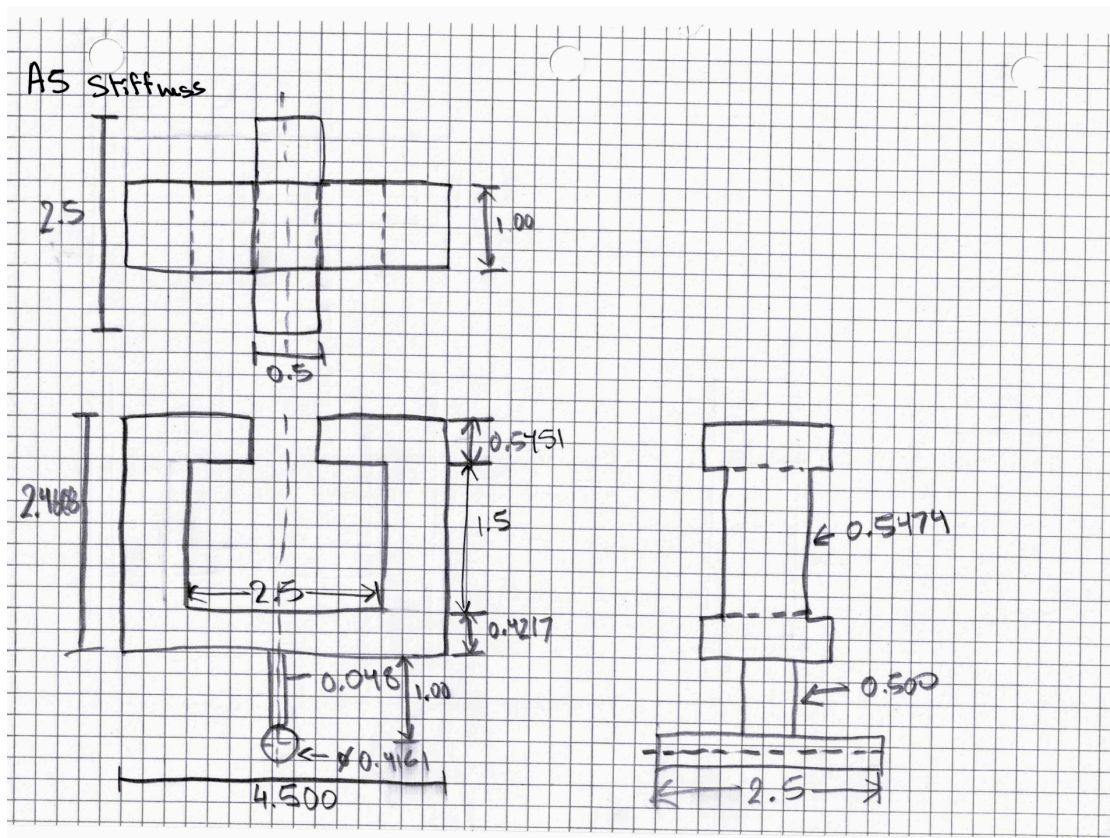
Strength governs all five calculated dimensions. D is the closest comparison: $0.6000 - 0.5474 = 0.0526$ in. Values in the calculation sketches are rounded analysis results; they are not manufacturing lower limits. The selected dimensions provide rounding margin.

Two hand-drawn multiview sketches

These are separate strength-only and stiffness-only sizing sketches, each containing front, top, and right-side views. Neither represents the final combined design by itself. The pin projects 1.000 in on each side of the 0.500-in-deep centered B connector, for a 2.500-in total length.



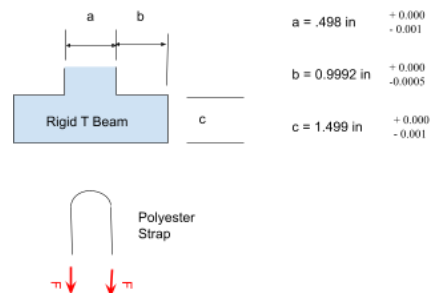
Hand-drawn strength multiview sketch; upper-body height = $1.500 + 0.7348 + 0.7348 = 2.9696$ in. .



Hand-drawn stiffness multiview sketch; upper-body height = $1.500 + 0.4217 + 0.5451 = 2.4668$ in. .

The strength sketch has D centerline height 2.2348 in; the stiffness sketch has 1.9834 in. Both are below the 2.250-in height used in D's original calculation. The stiffness drawing's small B-width label should read **0.0480 in**, consistent with the calculation. Any clipped overall-height label is supplied in the captions. The nominal openings shown on paper are supplemented by the fit limits below.

T-beam fits: a, b, and c



Assignment-provided T-beam dimensions and strap loading. .

The assignment's supplied external dimensions are $a = 0.4970\text{--}0.4980$, $b = 0.9987\text{--}0.9992$, and $c = 1.4980\text{--}1.4990$ in. The b lower deviation is **-0.0005 in**. Their intended classes are **RC7 free running**, **RC3 precision running**, and **RC4 close running**, respectively.

The selected handbook entries used for both the bracket and linkage are reproduced below as limit deviations in inches. Reference [1], printed pp. 634–635 and 640; table entries originally expressed in thousandths were divided by 1,000.

Use / class	Basic-size range (in)	Hole/internal deviation (in)	Shaft/external deviation (in)	Table / printed page
a / RC7	Over 0.40 to 0.71	0 to +0.0016	−0.0030 to −0.0020	4 / 635
b and link A / RC3	Over 0.71 to 1.19	0 to +0.0008	−0.0013 to −0.0008	3 / 634
c / RC4	Over 1.19 to 1.97	0 to +0.0016	−0.0020 to −0.0010	3 / 634
Link shaft / FN1	Over 0.95 to 1.19	0 to +0.0005	+0.0008 to +0.0012	9 / 640

Mating bracket dimension	Internal limits (in)	Minimum clearance (in)	Maximum clearance (in)
Stem slot a	0.5000–0.5016	0.5000 − 0.4980 = 0.0020	0.5016 − 0.4970 = 0.0046
Each flange recess b	1.0000–1.0008	1.0000 − 0.9992 = 0.0008	1.0008 − 0.9987 = 0.0021
Pocket height c	1.5000–1.5016	1.5000 − 1.4990 = 0.0010	1.5016 − 1.4980 = 0.0036

The recess b is measured horizontally from the corresponding stem-slot edge to the outer pocket wall. Thus full pocket width = $a + 2b = \mathbf{2.5000\text{--}2.5032\text{ in}}$. With 1.000-in sidewall widths, overall bracket width is 4.5000–4.5032 in. The selected upper-body height is $0.750 + c + 0.750 = 3.0000\text{--}3.0016\text{ in}$. These calculated limits supplement the nominal paper dimensions; no alteration of the supplied T-beam is required.

Manufacturing plan: mill the pocket and stem opening, finish the mating surfaces to the limits above, and inspect with calibrated internal measurements or gauges. Hold the common centerline and symmetric recess locations; size tolerances alone do not define alignment.

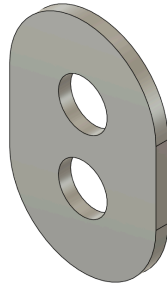
Final selected-member check including fit variation

At maximum opening sizes, use C span = 3.5032, E lever arm = 1.5008, D height = 2.2516, and $M = 600(1.5008) = 900.48\text{ lbf}\cdot\text{in}$. At A, use the smallest finished shaft diameter, 0.7487 in. These checks use the selected dimensions, not the smaller analysis-only sketches.

Feature	Maximum nominal normal stress (psi)	Local deformation (in)	Result
A, uniform strap load	7,281	0.000486	Below both limits
B	9,600	0.000960	Below both limits
C, with end couples	9,605	0.000890	Below both limits
D	6,003	0.002742	Below both limits
E	9,605	0.001923	Below both limits

The checks use the equations above with the revised lengths. For C, maximizing the same deflection polynomial with $L = 3.5032$ and $M_0 = 900.48$ gives the tabulated value. The largest stress is approximately 9,605 psi, giving nominal yield safety factor $40,000/9,605 = \mathbf{4.16}$. These are hand-analysis results, not FEA or test measurements.

MEGR 2157 linkage - strength and stiffness



Assignment-provided two-hole linkage concept. .

Known: $F = 600$ lbf per link, $E = 10^7$ psi, $\sigma_{\text{allow}} = 10,000$ psi. **Chosen:** width $w = 1.500$ minimum, hole-center distance $L = 3.000$, end radii 0.750 , overall length 4.500 . **Unknown:** thickness t , net areas, elongation, allowable center spacing, and hole/shaft limits. **Assumptions:** concentric tension and minimum net section between hole centers; same 0.005 -in deformation target. Two identical links preserve symmetry.

Strength at both holes

$$A_{\text{required}} = F/\sigma_{\text{allow}} = 600/10,000 = 0.0600 \text{ in}^2.$$

$$A_{\text{net}} = (w - d)t; \text{ use maximum permitted hole diameters.}$$

$$t_A = 0.0600/(1.500 - 0.7508) = \mathbf{0.08009 \text{ in.}}$$

$$t_{1\text{-in shaft}} = 0.0600/(1.500 - 1.0005) = \mathbf{0.12012 \text{ in.}}$$

Select **$t = 0.1250$ in minimum.**

$$A_A = (1.500 - 0.7508)(0.125) = 0.09365 \text{ in}^2.$$

$$A_1 = (1.500 - 1.0005)(0.125) = 0.0624375 \text{ in}^2.$$

$$\sigma_A = 600/0.09365 = \mathbf{6,407 \text{ psi.}}$$

$$\sigma_1 = 600/0.0624375 = \mathbf{9,610 \text{ psi.}}$$

$$n = 40,000/9,609.61 = \mathbf{4.16.}$$

Stiffness and allowable length

The minimum section is used over the complete 3.000 -in distance between centers, conservatively bounding the one-dimensional axial stretch. Local hole-contact compliance is excluded.

$$A_{\text{stiffness}} = FL/(E\delta_{\text{allow}}) = 600(3)/(10^7 \times 0.005) = \mathbf{0.0360 \text{ in}^2}.$$

$$t_{\text{stiffness}} = 0.0360/(1.500 - 1.0005) = \mathbf{0.07207 \text{ in.}}$$

$$\delta \leq FL/(EA_{\text{min}}) = 600(3)/(10^7 \times 0.0624375) = \mathbf{0.002883 \text{ in} < 0.005 \text{ in.}}$$

$$L_{\text{max}} = \delta_{\text{allow}}EA_{\text{min}}/F = 0.005(10^7)(0.0624375)/600 = \mathbf{5.203 \text{ in} > 3.000 \text{ in.}}$$

Manufacturing plan: machine the plate outline while maintaining minimum width and thickness; locate hole centers 3.000 in apart; drill undersize, then precision bore or ream to the limits. Turn and finish-grind the shafts. Inspect holes with bore gauges and shafts with micrometers, deburr, and provide small entry chamfers outside the working bearing surfaces. Slide the RC3 joint together. Support the plate around its lower hole and assemble the FN1 joint with an arbor press. These are planned processes; no part was manufactured or tested.

Verify the pin for the alternative linkage load

Locate each link center 0.500 in from its pin root. For this alternative FBD, replace A's uniform load with one 600-lbf downward force at $x = 0.500$ in; root reaction and moment remain 600 lbf and 300 lbf·in. Opposite links cancel their root moments at B.

$$M = Fa = 600(0.500) = 300 \text{ lbf·in.}$$

$$\sigma = 32M/(\pi d^3) = 32(300)/(\pi \times 0.7487^3) = \mathbf{7,281 \text{ psi.}}$$

$$I = \pi(0.7487)^4/64; L = 1.000; a = 0.500.$$

$$\delta_{\text{tip}} = Fa^2(3L - a)/(6EI) = \mathbf{0.000405 \text{ in.}}$$

Process, corrections, and lessons learned

Process and time. Approximately **5 hours** were spent on the assignment. The work proceeded from choosing material/load and drawing FBDs, through member sizing and link calculations, to separate paper multiview sketches and documentation.

Corrections recorded during review. Several handwritten moment arrows were reversed: A's root, C's left end, and D's bottom should be counterclockwise; E's root should be clockwise. These direction corrections are documented here alongside the original handwritten diagrams. The link's 9,610-psi result requires the actual 0.0624375-in² net area. An earlier drawing reference showed selected dimensions on both sheets; the completed paper sketches instead distinguish strength and stiffness sizing. The initial reading of b's lower tolerance was corrected to -0.0005 in, resolving the RC3 match. The original scans are retained; typed corrections do not alter the handwritten record.

Governing mode. Link strength requires 0.12012 in thickness while stiffness requires 0.07207 in, a difference of 0.04805 in. Strength governs, and the selected minimum thickness is 0.1250 in. For D, 0.6000 versus 0.5474 in is a closer comparison; it still favors strength.

Error propagation. E's reaction generates $600(1.500) = 900$ lbf·in, which passes through D into C. Omitting it would incorrectly leave C with only the center-load model. Checking each isolated FBD's force and moment balance caught the reversed arrow directions. The final check also carries the largest fit dimensions into the lever arms and member lengths instead of assuming that nominal dimensions are exact.

Assumption sensitivity. For A, a uniform 600-lbf load gives $M = WL/2 = 300$ lbf·in. If the same load acted at the free tip, M would be 600 lbf·in and the strength diameter would increase by $2^{1/3}$, from 0.6736 to approximately 0.8486 in. The 0.750-in selected pin would then need redesign. Unequal left/right loads would also introduce bending into B, invalidating its axial-only model. A detailed frame/contact analysis would be needed if connection flexibility, local stress concentrations, or interference-fit stresses were included.

References and original work

1. *Machinery's Handbook* excerpt, ANSI B4.1-1967 (R1987): Table 3, printed p. 634; Table 4, p. 635; Table 9, p. 640. Open the source tables. These are the printed pages in the accessible excerpt, not the assignment edition's pp. 646–660. The selected rows used in this design are shown above.
2. Kaiser Aluminum, *6061 Technical Data*, typical mechanical properties and physical properties. Open manufacturer data. Rounded $E = 10 \times 10^6$ psi and $S_y = 40$ ksi are used consistently in the hand model.
3. Updated course assignment, *Design for Strength and Stiffness I*, including the MEGR 2157 linkage requirements and appendices supplied with the assignment.

The handwritten work is embedded above. Unmodified originals are available for comparison: calculation scans and drawing scans.

Knowns:

$$F = 600 \text{ lbf}$$

$$\delta_{\text{allow}} = 0.005$$

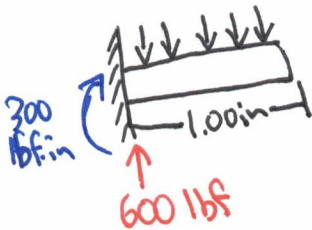
$$\sigma_{\text{allow}} = \frac{S_y}{SF} = \frac{40,000}{4} = 10,000$$

$$SF = 4$$

Material: 6061 T6 aluminum

$$E = 10,000,000 \text{ psi} \quad S_y = 40,000 \text{ psi}$$

Feature A:



Known:

$$W = 600 \text{ lbf}, \quad w = 600 \frac{\text{lbf}}{\text{in}}$$

$$L = 1.00 \text{ in}$$

$$M_{\text{max}} = \frac{WL}{2} = \frac{600(1)}{2} = 300 \text{ lbf}\cdot\text{in}$$

$$I = \frac{\pi d^4}{64}, \quad \sigma = \frac{M_c}{I} = \frac{32M}{\pi d^3}$$

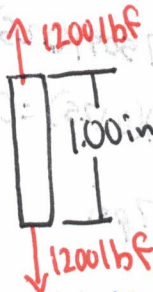
$$d_{\text{strength}} = \sqrt[3]{\frac{32M}{\pi \sigma_{\text{allow}}}} = \sqrt[3]{\frac{32(300)}{\pi (10,000)}} = 0.6$$

$$\delta = \frac{WL^2}{8EI} = \frac{8WL^3}{E\pi d^4}$$

$$\text{nominal } d = 0.750 \text{ in}$$

$$d_{\text{stiffness}} = \sqrt[4]{\frac{8WL^2}{E\pi \delta_{\text{allow}}}} = \sqrt[4]{\frac{8(600)(1)^2}{10^7 \pi (0.005)}} = 0.418 \text{ in}$$

Feature B:



Strength

$$A = bt, \quad \sigma = \frac{P}{bt}$$

Knowns

$$P = 1200 \text{ lbf}$$

$$b = 0.500 \text{ in}$$

$$L = 1.00 \text{ in}$$

unknown:

thickness: t

Stiffness:

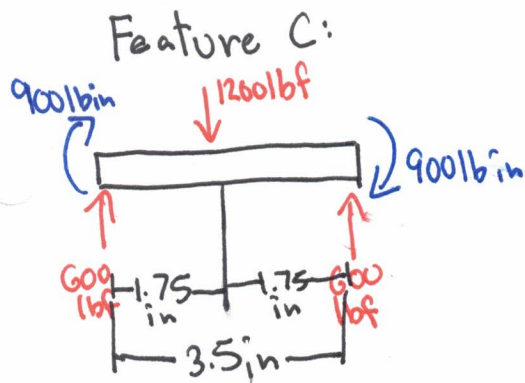
$$\delta = \frac{PL}{Eb^3t}$$

$$t_{\text{stiffness}} = \frac{PL}{Eb\delta_{\text{allow}}}$$

$$= \frac{1200(1)}{10^7(0.5)(0.005)} = 0.0480 \text{ in}$$

$$\text{select } t = 0.250 \text{ in}$$

$$t_{\text{strength}} = \frac{P}{b\sigma_{\text{allow}}} = \frac{1200}{0.5(10,000)} = 0.24 \text{ in}$$



Knowns:

$$P = 1200 \text{ lbf} \quad L = 3.50 \text{ in} \quad b = 1.00 \text{ in}$$

$$R_1 = R_2 = 600 \text{ lbf}$$

$$M_0 = 600(1.5) = 900 \text{ lbf}\cdot\text{in}$$

$$M(x) = -900 + 600x$$

$$I = \frac{bt^3}{12}, \quad \sigma = \frac{6M_{\max}}{bt^2}$$

$$M_{\max} = 900 \text{ lbf}\cdot\text{in}$$

$$t_{\text{strength}} = \sqrt{\frac{6M_{\max}}{b\sigma_{\text{allow}}}} = \sqrt{\frac{6(900)}{1(10000)}} = 0.7348 \text{ in}$$

$$EI_y = \frac{Px(3L^2 - 4x^2)}{48} - \frac{M_0x(L-x)}{2}$$

$$EI_y = -686.25x + 450x^2 - 100x^3$$

$$EI_y' = -686.25 + 900x - 300x^2 = 0$$

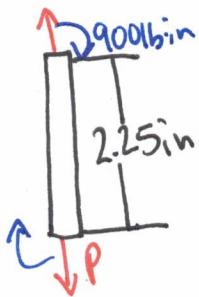
$$x = 1.25, 1.75 \text{ in}$$

$$\max EI_y = 312.5 \text{ lb}\cdot\text{in}^3$$

$$\delta_{\max} = \frac{312.5}{EI} = \frac{3750}{Ebt^3}$$

$$t_{\text{stiffness}} = \sqrt[3]{\frac{3750}{Eb\delta_{\text{allow}}}} = \sqrt[3]{\frac{3750}{10^7(1)(0.005)}} = 0.4217 \text{ in}$$

Feature D:



$$N = 600 \text{ lbf} \quad H = 2.25 \text{ in}$$

$$M = 900 \text{ lbf}\cdot\text{in} \quad h = 1.00 \text{ in}$$

$$A = ht, \quad I = \frac{th^3}{12}, \quad \sigma_{\max} = \frac{N}{ht} + \frac{6M}{th^2}$$

$$t_{\text{strength}} = \frac{N/h + \frac{6M}{h^2}}{\sigma_{\text{allow}}} = \frac{600 + \frac{6(900)}{1^2}}{10000} = 0.6000 \text{ in}$$

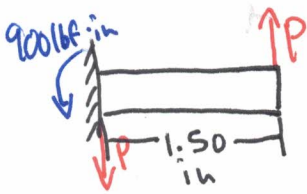
$$\delta_{\text{axial}} = \frac{NH}{Et} = \frac{600(2.25)}{E(1)} = 0.000135$$

$$\delta_{\text{lateral}} = \frac{MH^2}{2EI} = \frac{6MH^2}{Eth^3} = \frac{0.0027337}{1}$$

$$\delta = \sqrt{\delta_{\text{axial}}^2 + \delta_{\text{lateral}}^2}$$

$$t_{\text{stiffness}} = \frac{\sqrt{(0.000135)^2 + (0.0027337)^2}}{0.005} = 0.5474 \text{ in}$$

Feature E:



Known:

$$P = 600 \text{ lbf} \quad b = 1.00 \text{ in} \\ L = 1.5 \text{ in}$$

Unknown:

vertical thickness t

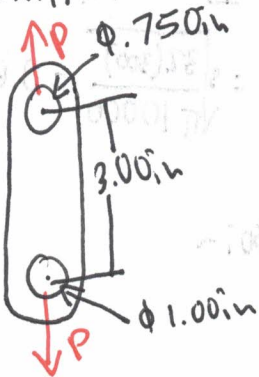
$$\sigma = \frac{6PL}{bt^2}$$

$$M_{\max} = PL = 600(1.5) = 900 \text{ lbf in}$$

$$t_{\text{strength}} = \sqrt{\frac{6PL}{b\sigma_{\text{allow}}}} = \sqrt{\frac{6(600)(1.5)}{1(10,000)}} = 0.7348 \text{ in}$$

$$t_{\text{stiffness}} = \sqrt[3]{\frac{4PL^3}{Eb\delta_{\text{allow}}}} = \sqrt[3]{\frac{4(600)(1.5)^3}{10^7(1)(0.005)}} = 0.5451 \text{ in}$$

Link:



$$P = 600 \text{ lbf} \quad w = 1.5 \text{ in}$$

$$L = 3.00 \text{ in}$$

$$L_{\text{overall}} = L + 2r = 3 + 2(0.75) = 4.5 \text{ in}$$

$$d_{\text{amax}} = 0.7508 \quad d_{\text{imin}} = 1.0005 \text{ in}$$

$$A_{\text{req}} = \frac{F}{\sigma_{\text{allow}}} = \frac{600}{10000} = 0.06 \text{ in}^2$$

$$A_{\text{net}} = (w - d)t$$

$$t_A = \frac{0.06}{1.5 - 0.7508} = 0.08009 \text{ in}$$

$$t_i = \frac{0.06}{1.5 - 1.0005} = 0.12012 \text{ in}$$

$$\sigma_i = \frac{600}{0.062} = 9610 \text{ psi}$$

$$\sigma_A = \frac{600}{0.09365} = 6407 \text{ psi}$$

$$n_{\min} = \frac{40000}{960961} = 4.16$$

Stiffness:

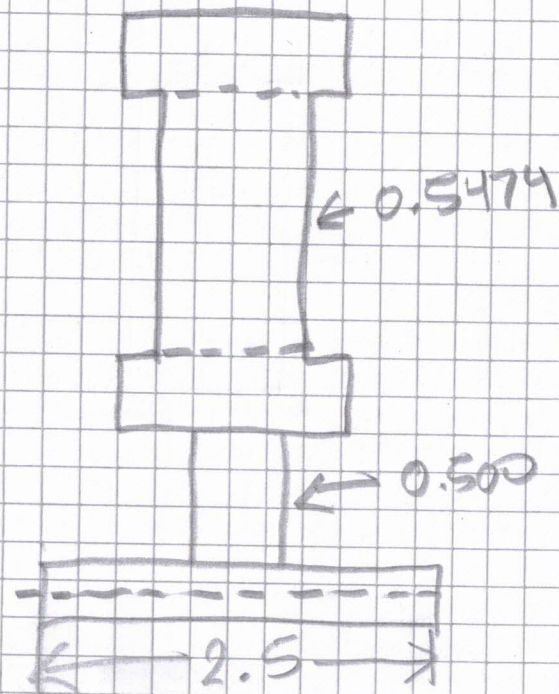
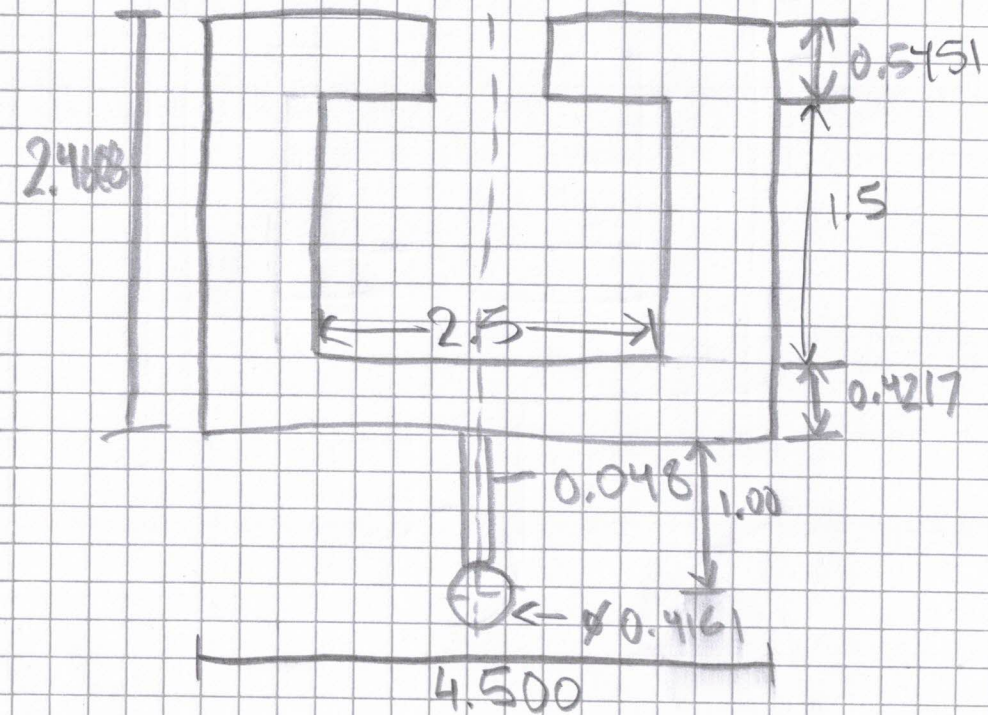
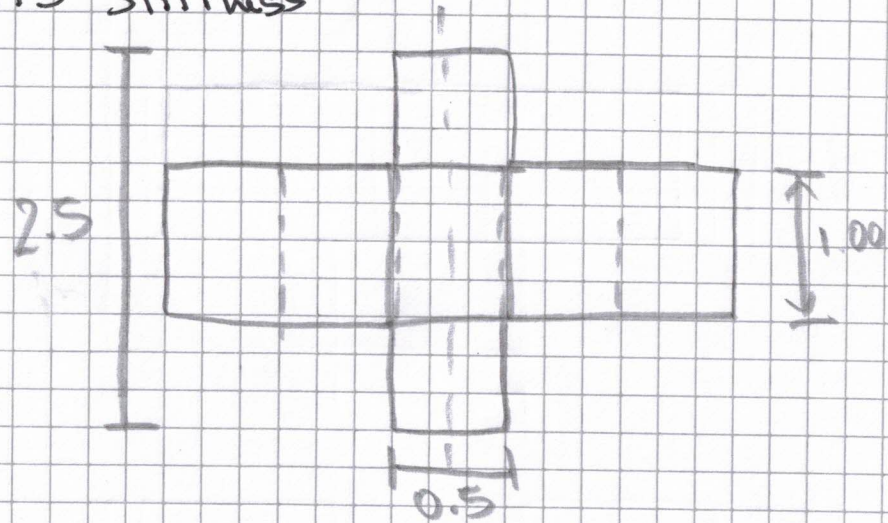
$$A_{\text{req}} = \frac{FL}{E\delta_{\text{allow}}} = \frac{600(3)}{10^7(0.005)} = 0.0360 \text{ in}^2$$

$$t_{\text{stiffness}} = \frac{0.036}{1.5 - 1.0005} = 0.07207 \text{ in}$$

$$\sum \frac{FL}{EA_{\min}} = \frac{600(3)}{10^7(0.0624)} = 0.002883$$

$$L_{\max} = \frac{\delta_{\text{allow}} EA_{\min}}{F} = \frac{0.005(10^7)(0.0624)}{600} = 5.203 \text{ in}$$

A5 Stiffness



AS strength

